Arithmetic obstructions in positive determinant approximations to $\beta(4)$

Research note prepared with Codex

31 October 2026

Abstract

A standard route to irrationality is to construct nonzero values of integer polynomials that are sufficiently small relative to their degrees. Positive integral representations offer a way to ensure nonvanishing, and determinants built from them can become very small. We examine two such constructions for $\beta(4) = 1 - 3^{-4} + 5^{-4} - \cdots$ and prove that the arithmetic cost defeats the real decay: after clearing denominators and removing every common coefficient factor, the polynomial values grow. For the first family this holds for every choice of consecutive positive integer nodes, even when their starting point grows arbitrarily with the matrix size. For the second we calculate exactly the unavoidable power of each sufficiently large prime in the normalization. The respective lower growth exponents are $2 - 2\log 2$ and $6 - \log 4$, on the scale of the square of the matrix size. These results identify why the two specified families fail as irrationality constructions. The proofs develop their positive integral representations, real estimates and prime-divisibility arguments; other determinant families remain outside the conclusions.

1 Introduction

Put

$$\theta = \beta(4) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^4}. \tag{1.1}$$

Positive integral representations provide nonzero quantities involving a constant. To see how this can help with irrationality, suppose that $\theta = a/b$ were rational, with $b > 0$. Every nonzero integer linear form $B\theta - A$ would then have absolute value at least $1/b$. Thus a sequence of nonzero integer linear forms tending to zero would contradict rationality. An accurate rational approximation A/B is useful for this purpose only when its error remains small after multiplication by B.

Determinant constructions extend this idea to polynomials. A determinant whose entries are rational linear forms in θ is the value $Q_n(\theta)$ of a rational polynomial, usually of degree n. If $Z_n \in \mathbb{Z}[X]$ has degree at most n, rationality would imply $b^n Z_n(\theta) \in \mathbb{Z}$. Consequently nonzero values with $|Z_n(\theta)|^{1/n} \to 0$ would suffice. Determinants can decay as $\exp(-cn^2)$, fast enough to overcome this degree cost, provided the passage from rational to integer coefficients does not consume the gain.

There are therefore two sizes to compare: the value of the rational determinant, and the multiplier needed to make its coefficients primitive integers. Here *primitive* means that their greatest common divisor is one. Write

$$Z_n = \kappa_n Q_n = \mathrm{Prim}(Q_n), \qquad \log|Z_n(\theta)| = \log\kappa_n + \log|Q_n(\theta)|. \tag{1.2}$$

The multiplier κ_n is positive and rational. Its determination includes both clearing denominators and cancelling common numerator factors. Ignoring the latter can give a false obstruction: an expensive chosen denominator need not be an unavoidable cost.

We prove that the actual cost is unavoidable in two explicit families. Their determinants are positive and become small, but their primitive integer-polynomial values grow exponentially on the n^2 scale. This answers a construction-level question: these families cannot supply the small values required above. It makes no assertion that θ is rational, or that other determinant designs must fail.

The relation between positive moments, determinant decay and irrationality criteria is developed systematically by Zudilin [6]. One motivation for studying $\beta(4)$ determinants is the mixed determinant strategy for Catalan's constant described by OpenAI [4, Section 1.2], which treats nonvanishing, the real estimate and prime valuations separately. The positive families considered here examine only part of the much wider space of determinant designs. Prévost [5] constructs positive-weight representations and Padé approximants for Hurwitz zeta values. Rational modifications of moment functionals belong to the classical orthogonal-polynomial framework; Krattenthaler [3] gives determinant formulas that include such modifications and repeated parameters. More directly, Carreira [2] investigates determinant constructions for $\beta(4)$, including positive functionals, primitive polynomial calculations and the competing real and arithmetic costs. Brown's discussion [1, Section 7.4] of denominator clearing, common divisors and congruences further emphasizes that the full coefficient content must be accounted for. We use these works to situate the question; no claimed global irrationality result in a recent manuscript is an input to our proofs.

The contributions here are specific uniform estimates and exact local arithmetic for the two families defined below. In each case we identify a coefficient that retains a large prime-power denominator after all cancellation. Combining these costs over many primes proves the obstruction. The first proof also shows that moving the entire block of consecutive nodes cannot repair that family. The second determines the cost at each large prime exactly.

1.1 Consecutive kernel determinants

The first construction seeks a symmetric matrix with rational off-diagonal entries and θ plus a rational number on each diagonal entry. We also want the matrix to be positive definite, so that its determinant cannot vanish. The following calculation explains the entries; positive definiteness is established separately in Section 3.

For integers $a \geq 0$ and $\ell \geq 2$, write

$$S_\ell(a) = \sum_{j=0}^{a-1} \left(\frac{1}{(4j+1)^\ell} - \frac{1}{(4j+3)^\ell} \right), \qquad S_\ell(0) = 0. \tag{1.3}$$

These are the first $2a$ terms of the alternating beta series of order ℓ, grouped into pairs. In particular, $\theta - S_4(a)$ is the remaining fourth-power tail. To obtain it by differentiation, start from the shifted cubic tail

$$f(t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1+4t)^3} \qquad (t \geq 0).$$

At a nonnegative integer a, shifting the summation index and differentiating give

$$f(0) - f(a) = S_3(a), \qquad f'(a) = -12\big(\theta - S_4(a)\big).$$

Thus subtracting the tail from $f(0)$ gives rational values at integer arguments, while its derivative introduces θ. Combine these properties by setting, for $x > 0$,

$$g(x) = \frac{\sqrt{x}}{6}\big(f(0) - f(\sqrt{x})\big), \qquad K(x,y) = \frac{g(y) - g(x)}{y - x}, \quad K(x,x) = g'(x).$$

Here K is a divided difference, with its limiting derivative on the diagonal. At squared integer nodes,

$$g(a^2) = \frac{aS_3(a)}{6}, \qquad g'(a^2) = \frac{S_3(a) - af'(a)}{12a} = \theta - S_4(a) + \frac{S_3(a)}{12a} \quad (a > 0).$$

The factor $\sqrt{x}$ compensates for the $1/(2\sqrt{x})$ from the chain rule, and the factor $1/6$ makes the coefficient of θ exactly one. These identities explain both the squared nodes and the constants in the matrix entries. For distinct positive integers $a_1, \ldots, a_n$, we have $[K(a_i^2, a_j^2)]_{i,j} = \theta I_n + R(a)$, where $R(a)$ is the rational symmetric matrix

$$R(a)_{ii} = -S_4(a_i) + \frac{S_3(a_i)}{12a_i}, \qquad R(a)_{ij} = \frac{a_j S_3(a_j) - a_i S_3(a_i)}{6(a_j^2 - a_i^2)} \quad (i \neq j). \tag{1.4}$$

The representation in Section 3 proves that K is a positive kernel: its matrices at distinct positive nodes are positive definite. This does not follow from the divided-difference formula alone. Replacing θ by X gives a *matrix pencil* $XI_n + R(a)$, whose entries depend linearly on X and whose determinant is monic. For consecutive nodes $a_i = m + i$, $1 \leq i \leq n$, set

$$Q_{n,m}(X) = \det(XI_n + R(a)), \qquad Z_{n,m}(X) = H_{n,m} Q_{n,m}(X) = \mathrm{Prim}(Q_{n,m})(X), \tag{1.5}$$

where $H_{n,m} > 0$ and $Z_{n,m} \in \mathbb{Z}[X]$ has coefficient gcd one.

Theorem 1.1 (Consecutive principal determinants)**.** *For every sequence of nonnegative integer offsets m_n, $Q_{n,m_n}(\theta) > 0$ and*

$$\liminf_{n\to\infty} \frac{\log Z_{n,m_n}(\theta)}{n^2} \geq 2 - 2\log 2 > 0. \tag{1.6}$$

The theorem allows arbitrary growth of m_n, including $m_n/n \to \infty$. The underlying kernel admits a uniform comparison for arbitrary positive integer row and column nodes (Theorem 3.3); the arithmetic conclusion above is restricted to consecutive principal nodes.

1.2 Difference-variable Gram determinants

The second construction builds positivity into the moments themselves:

$$\mu_m = \frac{1}{96} \int_0^1 \int_0^1 \frac{(u-v)^{2m} \log^2(uv)}{\sqrt{uv}(1+uv)} \, \mathrm{d}u \, \mathrm{d}v.$$

The weight is positive, and $y = (u-v)^2$ lies in $[0,1]$. Thus these are moments of a positive measure on that interval, and $[\mu_{i+j}]$ is the Gram matrix of $1, y, \ldots, y^{n-1}$. Its determinant is positive because a nonzero polynomial cannot vanish throughout the interval.

Why does this produce a polynomial in θ? Expand $(u-v)^{2m}$ and integrate its monomials. The middle term $u^m v^m$ contributes $\binom{2m}{m}(\theta - T_4(m))$; paired off-diagonal terms have rational values because their cubic beta tails cancel. The factor $1/96$ normalizes $\mu_0 = \theta$. Section 5 carries out this calculation and gives $\mu_m = C_m\theta + B_m$, with the explicit rational coefficients

$$T_\ell(j) = \sum_{k=0}^{j-1} \frac{(-1)^k}{(2k+1)^\ell}, \qquad C_m = \binom{2m}{m}, \tag{1.7}$$

$$B_m = -C_m T_4(m) + \frac{1}{6} \sum_{\substack{0 \leq j \leq 2m \\ j \neq m}} \binom{2m}{j} \frac{T_3(j)}{j-m}, \tag{1.8}$$

$$Q_n^\Delta(X) = \det[C_{i+j}X + B_{i+j}]_{0 \leq i,j < n}, \qquad Z_n^\Delta = \kappa_n Q_n^\Delta = \mathrm{Prim}(Q_n^\Delta), \quad \kappa_n > 0. \tag{1.9}$$

Here $T_\ell(j)$ truncates the beta series after j individual terms, so $S_\ell(a) = T_\ell(2a)$. The central binomial coefficient C_m comes from the middle term just described; B_m collects the finite tail corrections and the off-diagonal terms. Empty sums are zero. In particular $B_0 = 0$ and $B_1 = -149/81$.

Theorem 1.2 (Difference-variable Gram determinants)**.** *The polynomial Q_n^Δ has degree n and leading coefficient 2^{n-1}, and $Q_n^\Delta(\theta) > 0$. For every prime $p > 4n$,*

$$v_p(\kappa_n) = 3 \max \left\{ 0,\ 2n - 2 - \left\lfloor \frac{p}{4} \right\rfloor \right\}. \tag{1.10}$$

Moreover,

$$\log Z_n^\Delta(\theta) \geq (6 - \log 4)n^2 + o(n^2). \tag{1.11}$$

Both theorems imply that the corresponding primitive values eventually exceed one and that their nth roots tend to infinity. They exclude the specified sequences as small primitive values; they are not statements against all determinant methods for θ.

1.3 Proof organization and background

Both proofs follow the same three steps.

1. *Prove nonvanishing by positivity.* Write the determinant as an integral with positive integrand, or as the determinant of inner products for a positive measure. This avoids having to assume anything about the arithmetic nature of θ.
2. *Determine the scale of real decay.* Compare the determinant with an explicit Cauchy or moment determinant. An obstruction requires a lower bound: we must exclude the possibility that the real value is even smaller than an upper estimate suggests. The two-sided estimates give its logarithm to leading order.
3. *Prove an unavoidable arithmetic cost.* For selected primes, multiply by suitable powers of p and reduce the coefficients modulo p. A nonzero residue identifies a denominator that cannot cancel. Sum these costs using the prime number theorem, then use (1.2).

The following comparison shows the outcome before the details. The entries are coefficients of n^2, with errors $o(n^2)$ omitted. In the first row the nodes are $1, \dots, n$; the proof then treats every offset.

Family	$\log Q_n(\theta)$	$\log \kappa_n$ (lower)	$\log Z_n(\theta)$ (lower)
Consecutive, $m = 0$	$-2\log 2$	2	$2 - 2\log 2$
Difference Gram	$-\log 4$	6	$6 - \log 4$

The positive final column is the obstruction. The estimates also explain why the arithmetic calculation must account for all common coefficient factors: otherwise that column would describe only one normalization.

Section 2 gives the elementary normalization facts. Sections 3–4 carry out the three steps for the consecutive family; Sections 5–6 do so for the difference Gram family. Their arguments can be read separately after the normalization section. Section 7 describes the scope and evidence, and Appendix A extends the first construction to include a zero node.

We use elementary complex integration, linear algebra, Stirling's formula and the prime number theorem in the form

$$\vartheta(x) := \sum_{p \leq x} \log p = x + o(x). \tag{1.12}$$

The latter is the only theorem on prime distribution used. The specific kernel identities, determinant comparisons, moment formulas and local valuation arguments are proved below. No numerical optimization or finite enumeration is an input to either asymptotic obstruction.

2 Primitive normalization and small polynomial values

Definition 2.1. For $0 \neq Q \in \mathbb{Q}[X]$, its primitive representative $\mathrm{Prim}(Q)$ is the unique positive rational multiple $\kappa Q \in \mathbb{Z}[X]$ whose coefficients have gcd one. This convention preserves the sign of Q.

We write $[X^j]Q$ for the coefficient of X^j in Q. For a nonzero rational number, v_p counts powers of p in its numerator minus those in its denominator; set $v_p(0) = +\infty$. For example $v_p(p^{-3}) = -3$. Saying that a rational number is p-integral means that its denominator in lowest terms is not divisible by p. Reduction modulo p below is always performed on p-integral quantities, by inverting the denominator modulo p. A p-unit has valuation zero, so its reduction is nonzero.

Lemma 2.2. *If $Q(X) = \sum_j q_j X^j \neq 0$ and $\mathrm{Prim}(Q) = \kappa Q$, then*

$$v_p(\kappa) = -\min_j v_p(q_j) \tag{2.1}$$

for every prime p. If Q is monic, κ is a positive integer. If the leading coefficient of Q is 2^e with $e \geq 0$, then $v_p(\kappa) \geq 0$ for odd p and $v_2(\kappa) \geq -e$.

Proof. Integral primitive coefficients have minimum valuation zero at every prime, which proves (2.1). A monic polynomial has a coefficient of valuation zero at every prime. A leading coefficient 2^e has valuation zero at odd primes and valuation e at 2. □

A sufficient common denominator can overstate the primitive multiplier. For example, $Q(X) = (6X + 12)/5$ becomes integral after multiplication by 5, but $\mathrm{Prim}(Q) = X + 2$ and its primitive multiplier is $5/6$. A lower bound on a selected denominator-clearing integer therefore need not be a lower bound on the primitive multiplier. Formula (2.1) is the distinction used in both main proofs.

If U, V are invertible rational matrices, then $\det(UM(X)V) = \det U \det V \det M(X)$. Consequently the primitive polynomial of a full pencil is unchanged up to sign by rational invertible row and column operations. Passing to a different subspace or a different pencil is a different construction and is not covered by this observation.

For completeness, if $\theta = a/b \in \mathbb{Q}$ with $b \geq 1$ and $Z \in \mathbb{Z}[X]$ has degree at most n, then $b^n Z(\theta) \in \mathbb{Z}$. Thus nonzero values with $|Z_n(\theta)|^{1/n} \to 0$ would contradict rationality. The degree factor is essential: merely having $Z_n(\theta) \to 0$ with increasing degree need not do so. For example, $Z_n(X) = X^n$ has value 2^{-n} at the rational number $1/2$. The main theorems give growth in the opposite direction for their two specified families.

3 The positive cubic-tail kernel

The introduction explains the rational entries of the kernel matrix. The remaining task is to prove positivity and estimate its determinant. We first express $K(x, y)$ as an integral of $(x+s^2)^{-1}(y+s^2)^{-1}$ against a positive density. These factors make a Gram representation when the two node lists agree, and an explicit determinant integral handles different row and column lists. We then compare that density with simpler ones that yield Cauchy determinants. This separates the positivity argument from the size estimate.

3.1 A direct integral representation

For $\mathrm{Re}\, z \geq 0$, define

$$f(z) = \sum_{j=0}^{\infty} \left((4j + 1 + 4z)^{-3} - (4j + 3 + 4z)^{-3} \right). \tag{3.1}$$

Expanding $\operatorname{sech} u = 2\sum_{k\ge 0}(-1)^k e^{-(2k+1)u}$ and integrating gives

$$f(z) = \frac{1}{4}\int_0^\infty u^2 \operatorname{sech}(u) e^{-4zu}\,\mathrm{d}u, \quad f(a) = f(0) - S_3(a), \quad f'(a) = -12(\theta - S_4(a)) \tag{3.2}$$

at nonnegative integer a. The integrated series is absolutely summable, which justifies the expansion and differentiation used here.

The next two transforms convert the exponential in this Laplace integral into the rational factors needed for the kernel. The cosine transform evaluates the resulting density explicitly; the second identity replaces $\sqrt{x}(1 - e^{-t\sqrt{x}})$ by an integral containing $x/(x + s^2)$. We include their proofs to fix the normalization:

$$\int_0^\infty \operatorname{sech}(u)\cos(tu)\,\mathrm{d}u = \frac{\pi}{2}\operatorname{sech}\frac{\pi t}{2}, \tag{3.3}$$

$$\int_0^\infty \frac{x(1-\cos(ts))}{x+s^2}\,\mathrm{d}s = \frac{\pi\sqrt{x}}{2}(1 - e^{-t\sqrt{x}}), \qquad x > 0,\ t \ge 0. \tag{3.4}$$

For (3.3), integrate $e^{itz}/\cosh z$ around the rectangle with vertices $\pm R, \pm R + i\pi$. The top edge contributes $e^{-\pi t}$ times the bottom edge, and the pole at $i\pi/2$ has residue $e^{-\pi t/2}/i$. The vertical edges tend to zero. Hence the full-line transform is $\pi \operatorname{sech}(\pi t/2)$; symmetry gives the displayed formula. For (3.4), the upper-half-plane pole at $i\sqrt{x}$ gives $\int_0^\infty \cos(ts)/(s^2 + x)\,\mathrm{d}s = \pi e^{-t\sqrt{x}}/(2\sqrt{x})$ for $t > 0$, and the value at $t = 0$ follows by continuity. Subtract the latter transforms and multiply by x.

Differentiating (3.3) twice under the integral gives

$$f(0) = \frac{\pi^3}{32}, \qquad \operatorname{Re} f(is) = \frac{\pi^3}{32}A(s)(2A(s)^2 - 1), \quad A(s) = \operatorname{sech}(2\pi s). \tag{3.5}$$

Define

$$g(x) = \frac{\sqrt{x}}{6}(f(0) - f(\sqrt{x})), \quad g(0) = 0, \qquad K(x,y) = \frac{g(y) - g(x)}{y - x}, \tag{3.6}$$

with the derivative value when $x = y$.

Proposition 3.1. *For $x, y \ge 0$,*

$$K(x,y) = \int_0^\infty \frac{\omega(s)}{(x+s^2)(y+s^2)}\,\mathrm{d}s, \quad \omega(s) = \frac{\pi^2}{96}s^2 q(s), \quad q(s) = 1 + A(s) - 2A(s)^3. \tag{3.7}$$

The density ω is strictly positive for $s > 0$. On squared integer nodes the kernel equals

$$K(a^2, b^2) = \frac{bS_3(b) - aS_3(a)}{6(b^2 - a^2)} \qquad (a \ne b), \tag{3.8}$$

$$K(a^2, a^2) = \theta - S_4(a) + \frac{S_3(a)}{12a} \qquad (a > 0), \tag{3.9}$$

$$K(0,0) = 2\theta. \tag{3.10}$$

Every square minor with strictly increasing nonnegative row nodes and strictly increasing nonnegative column nodes is positive.

Proof. First identify the density. Equations (3.2) and (3.4), with $t = 4u$, give

$$g(x) = \frac{1}{12\pi}\int_0^\infty \int_0^\infty u^2 \operatorname{sech} u \,\frac{x(1-\cos(4us))}{x+s^2}\,\mathrm{d}s\,\mathrm{d}u.$$

The integrand is nonnegative, so Tonelli's theorem permits interchange. The inner u-integral is $4(f(0) - \operatorname{Re} f(is))$. Hence

$$g(x) = \int_0^\infty \frac{x}{x+s^2}\sigma(s)\,\mathrm{d}s, \qquad \sigma(s) = \frac{f(0) - \operatorname{Re} f(is)}{3\pi} = \frac{\pi^2}{96}q(s). \tag{3.11}$$

Taking a divided difference uses the elementary identity

$$\frac{y/(y+s^2) - x/(x+s^2)}{y-x} = \frac{s^2}{(x+s^2)(y+s^2)},$$

which gives (3.7). Positivity is now explicit:

$$q = (1-A)(2A^2+2A+1) > 0 \qquad (s>0). \tag{3.12}$$

Since $\omega(s) = O(s^4)$ at zero and $O(s^2)$ at infinity, the integral is finite even when $x = y = 0$. Differentiation or dominated convergence justifies the diagonal and zero limits. The identities (3.8)–(3.9) follow from (3.2); at zero, $g(x) = -f'(0)x/6 + O(x^{3/2}) = 2\theta x + O(x^{3/2})$.

It remains to pass from the kernel to its minors. For increasing nodes x_i, y_j, expanding the two determinants and integrating termwise gives the determinant integration identity

$$\det[K(x_i,y_j)]_{i,j=1}^n = \frac{1}{n!}\int_{(0,\infty)^n} \det\left[\frac{1}{x_i+s_\ell^2}\right]_{i,\ell} \det\left[\frac{1}{y_j+s_\ell^2}\right]_{j,\ell} \prod_{\ell=1}^n \omega(s_\ell)\,\mathrm{d}s_1\cdots\mathrm{d}s_n. \tag{3.13}$$

Absolute integrability follows by expanding into finitely many products of the finite one-variable integrals. This is the usual Andréief–Heine identity used in positive determinant arguments [6, Section 2], with its elementary expansion proof here. For $0 < s_1 < \cdots < s_n$, both Cauchy determinants are positive by the Cauchy product formula (3.14), applied to x_i, s_j^2 and y_i, s_j^2. Their product is unchanged by permuting the s_ℓ and is strictly positive away from coincidences. This proves the last assertion. It also proves that increasing the density increases these determinants, even for different row and column nodes. □

Remark 3.2. The density in (3.7) satisfies $\omega(s) \sim \pi^2 s^2/96$ at infinity; it has infinite total mass, but the two resolvent factors make every kernel entry integrable. Carreira's functional [2, Definition 24(B) and Lemma 25] uses a positive density obtained from derivatives of $1/\sinh(\pi s)$, with exponential decay and finite polynomial moments. The representation used here is derived directly, and assumes no polynomial moments of $\omega(s)\,\mathrm{d}s$.

3.2 A uniform comparison with Cauchy determinants

The positive representation makes nonvanishing transparent, but its density is inconvenient for estimating a large determinant. We compare that density with simpler ones whose determinants have explicit product formulas. The comparison changes the logarithm by only $O(n)$, so it preserves the n^2 scale relevant to the obstruction. Positivity of the determinant integrand is what makes this comparison legitimate.

For positive nodes a_i, b_j, write

$$C(a,b)_{ij} = \frac{1}{a_i+b_j}, \qquad \det C(a,b) = \frac{\prod_{i<j}(a_j-a_i)(b_j-b_i)}{\prod_{i,j}(a_i+b_j)}. \tag{3.14}$$

The product formula follows, for example, by multiplying the determinant by its displayed denominator: the numerator is alternating separately in the two node sets, hence contains their

Vandermonde factors, and comparison of degrees and one residue fixes the remaining constant to one. The constants below have three roles: M bounds the density factor q, τ_0 fixes a lower comparison that vanishes quadratically at zero, and c_0 includes the factor $\pi/2$ from integrating the comparison kernel. Set

$$c_0 = \frac{\pi^3}{192}, \qquad M = 1 + \frac{2}{3\sqrt{6}}, \qquad \tau_0 = \frac{1}{\sqrt{2}\pi}. \tag{3.15}$$

Theorem 3.3 (Uniform Cauchy comparison)**.** *For any two strictly increasing lists of positive integers a_i, b_j,*

$$\left(\frac{c_0}{(1+\tau_0)^2}\right)^n \det C(a,b) \le \det[K(a_i^2, b_j^2)] \le (Mc_0)^n \det C(a,b). \tag{3.16}$$

In particular,

$$\log\det[K(a_i^2, b_j^2)] = \log\det C(a,b) + O(n), \tag{3.17}$$

with an absolute constant independent of all node positions and gaps.

Proof. The maximum of $1 + A - 2A^3$ for $0 \le A \le 1$ is M. Also $\cosh v \ge 1 + v^2/2$ and (3.12) imply

$$\frac{s^2}{s^2+\tau_0^2} \le 1 - \operatorname{sech}(2\pi s) \le q(s) \le M. \tag{3.18}$$

The elementary rational integrals needed for the comparison are

$$\int_0^\infty \frac{s^2\,\mathrm{d}s}{(s^2+a^2)(s^2+b^2)} = \frac{\pi}{2(a+b)}, \tag{3.19}$$

$$\int_0^\infty \frac{s^4\,\mathrm{d}s}{(s^2+\tau^2)(s^2+a^2)(s^2+b^2)} = \frac{\pi}{2} R_\tau(a,b), \tag{3.20}$$

$$R_\tau(a,b) = \frac{ab + \tau(a+b)}{(a+b)(a+\tau)(b+\tau)}. \tag{3.21}$$

To check the second formula, first divide $s^4/((s^2+a^2)(s^2+b^2))$ into $[a^2s^2/(s^2+a^2) - b^2s^2/(s^2+b^2)]/(a^2-b^2)$, and then use the first formula with the extra factor $s^2+\tau^2$. Coincident parameters follow by continuity. The first formula follows by partial fractions and $\int_0^\infty (s^2+a^2)^{-1}\,\mathrm{d}s = \pi/(2a)$.

Applying the positive determinant integral (3.13) to the density bounds gives

$$c_0^n \det[R_{\tau_0}(a_i,b_j)] \le \det[K(a_i^2,b_j^2)] \le (Mc_0)^n \det C(a,b). \tag{3.22}$$

This uses the sign of the determinant integrand; an entrywise matrix comparison alone would not suffice.

We now evaluate the regularized determinant exactly:

$$\begin{aligned}\det[R_\tau(a_i,b_j)] = {} & \det C(a,b) \prod_i \frac{a_i}{a_i+\tau} \prod_j \frac{b_j}{b_j+\tau} \\ & \times \left(1 + \tau\sum_i \frac{1}{a_i} + \tau \sum_j \frac{1}{b_j}\right).\end{aligned} \tag{3.23}$$

Let $u_i = 1/a_i$, $v_j = 1/b_j$. The entrywise factorization

$$R_\tau(a_i,b_j) = \frac{a_i}{a_i+\tau}\left(\frac{1}{a_i+b_j} + \frac{\tau}{a_i b_j}\right)\frac{b_j}{b_j+\tau}$$

reduces the calculation to $\det(C+\tau uv^t)$. The rank-one identity $\det(C+\tau uv^t) = \det C(1+\tau v^t C^{-1}u)$ leaves one scalar to compute. If $w = C^{-1}u$ and $F(x) = \sum_j w_j/(x+b_j)$, then $F(a_i) = 1/a_i$. Therefore

$$\frac{1}{x} - F(x) = \frac{1}{x}\,\frac{\prod_i(1-x/a_i)}{\prod_j(1+x/b_j)}.$$

Indeed, after multiplication by $x\prod_j(x+b_j)$, the numerator has degree at most n and vanishes at every a_i; its factor is fixed by the residue one at zero. Comparing constant terms at zero gives $v^tC^{-1}u = F(0) = \sum_i 1/a_i + \sum_j 1/b_j$. The rank-one determinant identity proves (3.23). Finally each integer node is at least one, so the products are at least $(1+\tau_0)^{-2n}$ and the last factor is at least one. This proves (3.16) and (3.17). □

4 The consecutive principal obstruction

We now determine part of the arithmetic cost in (1.2). For each prime in a suitable interval, the fourth-power tails contribute a diagonal residue that the cubic tails cannot cancel. A coefficient of the determinant records this residue. One such coefficient per prime is enough for a lower bound on the actual primitive multiplier.

4.1 A coefficient with forced valuation

For an odd prime p put $\chi_4(p) = (-1)^{(p-1)/2}$. Writing M_0 for the largest node, the interval $2M_0 < p < 4M_0$ below is chosen to isolate a single denominator p in each beta partial sum, while keeping all factors from differences of nodes invertible modulo p. Multiplication by p^4 then kills the cubic terms and detects only the fourth-power diagonal terms. We call a_i a high node when $4a_i > p$.

Lemma 4.1. *Let $a_1 < \cdots < a_n$ be positive integers with maximum M_0, and set $Q(X) = \det(XI_n + R(a))$. If $p \geq 5$ and $2M_0 < p < 4M_0$, put $r_p = \#\{i : 4a_i > p\}$. Then*

$$v_p([X^{n-r_p}]Q) = -4r_p, \qquad v_p(\kappa(Q)) \geq 4r_p, \tag{4.1}$$

where $\mathrm{Prim}(Q) = \kappa(Q)Q$.

Proof. The quantities a_i, $a_i + a_j$ and $a_i - a_j$ for $i \neq j$ are p-units. Among odd denominators below $4M_0$, the only possible multiple of p is p itself, since $3p > 4M_0$. It appears in $S_\ell(a_i)$ exactly when $4a_i > p$, with sign $\chi_4(p)$. Consequently $p^4R(a)$ is p-integral and its reduction is diagonal:

$$p^4R(a)_{ij} \equiv \begin{cases} -\chi_4(p), & i = j,\ 4a_i > p, \\ 0, & \text{otherwise} \end{cases} \pmod p. \tag{4.2}$$

Expanding $\det(XI + R)$ by the set of diagonal entries from which X is selected shows that the coefficient of X^{n-r_p} is the sum of all principal minors of R of size r_p. After multiplication by p^{4r_p}, exactly one such minor survives modulo p: that using all the high nodes. Its residue is $(-\chi_4(p))^{r_p} \neq 0$. For $r_p = 0$ this is the leading coefficient one. This proves the equality in (4.1); Lemma 2.2 proves the inequality for the actual primitive multiplier. □

Remark 4.2 (A three-node example). Take nodes $1, 2, 3$ and $p = 7$. The high nodes in the lemma are $2, 3$, and $\chi_4(7) = -1$, so

$$7^4R \equiv \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \pmod 7.$$

The coefficient of X in $\det(XI_3 + R)$ is the sum of the three principal 2×2 minors:

$$[X]Q = (R_{11}R_{22} - R_{12}^2) + (R_{11}R_{33} - R_{13}^2) + (R_{22}R_{33} - R_{23}^2).$$

After multiplication by 7^8, only the final minor survives modulo 7, with residue 1. Thus $v_7([X]Q) = -8$, and the primitive multiplier must contain at least 7^8. This is the role of the uniquely surviving minor: it rules out cancellation in the sum defining that coefficient. Other coefficients may force additional powers; a lower bound is enough.

Since Q is monic, the primitive multiplier is integral. Thus the local inequalities can be summed without any negative contribution from omitted primes:

$$\log \kappa(Q) \geq 4 \sum_{2M_0 < p < 4M_0} r_p \log p \tag{4.3}$$

for M_0 sufficiently large. This is a lower bound after all coefficient content has been removed.

4.2 Proportional offsets

We now compare two functions of the relative offset m/n: $\mathcal{A}$ measures the forced denominator cost, and $\mathcal{D}$ measures the real decay. Their difference will be the lower bound for the primitive value. This subsection treats finite limiting offsets; the next handles $m/n \to \infty$ separately so that no uniformity in an unbounded parameter is assumed.

Take $a_i = m + i$, $M_0 = m + n$, and suppose first that $m/n \to \rho < \infty$. Reverse the finite sum in (4.3). Since the endpoints $2M_0$ and $4a_i$ are even and exceed two, endpoint conventions for primes have no effect. The prime number theorem, uniformly for $2M_0 \leq x \leq 4M_0$, gives

$$\begin{aligned} 4 \sum_{2M_0 < p < 4M_0} r_p \log p &= 4 \sum_{a_i > M_0/2} \left(\vartheta(4a_i) - \vartheta(2M_0)\right) \\ &= 16 \sum_{a_i > M_0/2} (a_i - M_0/2) + o(n^2) \\ &= \mathcal{A}(\rho) n^2 + o(n^2), \end{aligned} \tag{4.4}$$

where

$$\mathcal{A}(\rho) = \begin{cases} 2(1+\rho)^2, & 0 \leq \rho \leq 1, \\ 8\rho, & \rho \geq 1. \end{cases} \tag{4.5}$$

Reversing the prime sum counts, for each node a_i, the primes from $2M_0$ up to $4a_i$. With $a_i/n \to \rho + t$, the resulting Riemann sum is $16 \int_0^1 (\rho + t - (\rho+1)/2)_+ \, \mathrm{d}t$, where $u_+ = \max(u, 0)$. When $\rho \leq 1$, only the upper part of the node interval contributes; when $\rho \geq 1$, every node contributes. These are the two cases in (4.5). Uniformity of the prime-number-theorem error follows from $\sup_{2M_0 \leq x \leq 4M_0} |\vartheta(x) - x| = o(M_0)$ and $M_0 = O(n)$.

For the real estimate, the differences $a_j - a_i = j - i$ do not depend on the offset. The Vandermonde product is $\prod_{i<j}(j-i) = \prod_{j=1}^{n-1} j!$, so the Cauchy product is

$$\det C(a, a) = \frac{\prod_{j=1}^{n-1} (j!)^2}{\prod_{i,j=1}^{n} (2m + i + j)}. \tag{4.6}$$

Stirling's formula, followed by a Riemann sum for the denominator, gives

$$\log \det C(a, a) = -\mathcal{D}(\rho) n^2 + o(n^2), \qquad \mathcal{D}(\rho) = \frac{3}{2} + \int_0^1 \int_0^1 \log(2\rho + x + y) \, \mathrm{d}x \, \mathrm{d}y. \tag{4.7}$$

For clarity, $2\sum_{j<n}\log(j!) = n^2\log n - \frac{3}{2}n^2 + O(n\log(n+1))$. At $\rho = 0$ the logarithmic corner singularity is integrable and its Riemann sums converge. Indeed, counting at most k grid points with $i+j=k$ bounds the contribution from $i+j \le \varepsilon n$ by $O(\varepsilon^2(1+|\log\varepsilon|)) + O(\log(n+1)/n)$. A positive offset tending to zero is handled by monotonicity between offsets zero and any fixed $\varepsilon > 0$. Integrating explicitly yields

$$\begin{aligned}\mathcal{D}(\rho) = \frac{1}{2}\big((2\rho+2)^2\log(2\rho+2) - 2(2\rho+1)^2\log(2\rho+1) \\ + (2\rho)^2\log(2\rho)\big),\end{aligned} \tag{4.8}$$

where $0^2\log 0 = 0$.

It remains to show that shifting cannot make real decay overtake the arithmetic cost. Differentiating the integral formula shows that the derivative of $\mathcal{D}$ is decreasing, and

$$\mathcal{D}'(\rho) = 2\int_0^1\int_0^1 \frac{\mathrm{d}x\,\mathrm{d}y}{2\rho+x+y} \le \mathcal{D}'(0) = 4\log 2 < 4. \tag{4.9}$$

On both pieces of (4.5), $\mathcal{A}'(\rho) \ge 4$. It follows that

$$\mathcal{A}(\rho) - \mathcal{D}(\rho) \ge \mathcal{A}(0) - \mathcal{D}(0) = 2 - 2\log 2. \tag{4.10}$$

Combining (4.3), (4.4), (4.7) and Theorem 3.3 proves the stronger proportional-offset statement

$$\log Z_{n,m}(\theta) \ge (\mathcal{A}(\rho) - \mathcal{D}(\rho))n^2 + o(n^2). \tag{4.11}$$

4.3 Offsets larger than the matrix size

Suppose now that $m/n \to \infty$. All nodes exceed $M_0/2$ eventually. The relevant scale is now mn: the prime cost is of this order, whereas the logarithm of the real determinant has size at most $O(n^2\log(1+m/n)) = o(mn)$. The same reversal of sums gives

$$\begin{aligned}\log H_{n,m} &\ge 16\sum_{i=1}^{n}(m+i-(m+n)/2) + o(nM_0) \\ &= 8mn + 8n + o(mn).\end{aligned} \tag{4.12}$$

The relative prime-number-theorem error is still uniform on $[2M_0, 4M_0]$, so this argument imposes no upper bound on m. By (4.6) and the numerator estimate just used,

$$\log\det C(a,a) \ge -n^2\big(\tfrac{3}{2} + \log(2(1+m/n))\big) - O(n\log(n+1)) = -o(mn). \tag{4.13}$$

The Cauchy comparison costs only $O(n)$, whence

$$\log Z_{n,m}(\theta) \ge (8 - o(1))mn. \tag{4.14}$$

Every sequence of ratios m_n/n has subsequences converging in the compactified interval $[0,\infty]$. If the liminf asserted in Theorem 1.1 failed, a subsequence witnessing that failure would have a further subsequence with such a limit. Equations (4.11)–(4.14) exclude every possible limit. Positivity follows from Proposition 3.1. This completes the proof of Theorem 1.1.

5 Difference-variable Gram determinants

The second design begins with a positive double integral. Its monomial integrals involve fourth-power beta tails on the diagonal and cubic tails off the diagonal. Expanding an even power of $u - v$ makes the unwanted $\beta(3)$ terms cancel, leaving rational linear forms in θ. Moreover $y = (u - v)^2$ lies in $[0, 1]$, so its powers form a positive moment sequence on a short interval. This gives both a nonvanishing Gram determinant and an effective estimate of its real size.

5.1 Moment formula and target isolation

For a polynomial $P(u, v)$, define the finite integral

$$J(P) = \int_0^1 \int_0^1 \frac{P(u,v)\log^2(uv)}{\sqrt{uv}(1+uv)}\,\mathrm{d}u\,\mathrm{d}v, \qquad \mu_m = \frac{1}{96}J((u-v)^{2m}). \tag{5.1}$$

The weight is integrable and strictly positive on the open square.

Lemma 5.1. *With C_m, B_m defined in (1.7)–(1.8),*

$$\mu_m = C_m\theta + B_m. \tag{5.2}$$

Proof. First compute the integral of one monomial; then expand $(u - v)^{2m}$. The identity

$$\int_0^1 \int_0^1 u^{a+t-1}v^{b+t-1}\,\mathrm{d}u\,\mathrm{d}v = \frac{1}{(a+t)(b+t)}$$

shows why two common-parameter derivatives insert $\log^2(uv)$. Expand $(1 + uv)^{-1}$ as a geometric series. For $a, b > 0$, the log-squared integral in each term is therefore the second derivative of the right side at $t = 0$. For $a \neq b$ this derivative is

$$\frac{2}{b-a}\left(\frac{1}{a^3} - \frac{1}{b^3}\right),$$

and for $a = b$ it is $6/a^4$. The sum of the absolute integrals of the terms is finite (their tails are $O(k^{-4})$), justifying termwise integration. Substitution of $a = k + i + 1/2$, $b = k + j + 1/2$ gives

$$\frac{J(u^iv^i)}{96} = (-1)^i(\theta - T_4(i)), \tag{5.3}$$

$$\frac{J(u^iv^j)}{96} = -\frac{(-1)^i}{6}\frac{T_3(i) - T_3(j)}{j-i} \quad (i \neq j,\ i + j \text{ even}). \tag{5.4}$$

In (5.4), the two cubic tails have the same sign $(-1)^i = (-1)^j$, so their $\beta(3)$ terms cancel. Now expand $\mu_m = \sum_{j=0}^{2m}(-1)^j\binom{2m}{j}J(u^jv^{2m-j})/96$. Its diagonal term contributes $\binom{2m}{m}(\theta - T_4(m))$ by (5.3). For $j \neq m$, the sign $(-1)^j$ cancels the corresponding sign in (5.4). The off-diagonal sum becomes

$$-\frac{1}{12}\sum_{\substack{0\le j\le 2m\\ j\neq m}}\binom{2m}{j}\frac{T_3(j) - T_3(2m-j)}{m-j}.$$

Changing j to $2m - j$ in the second part makes the two parts equal. The result is the sum in (1.8), proving (5.2). □

All coefficient statements use the explicitly defined polynomial $C_mX + B_m$. No independence or irrationality of θ is assumed to identify its coefficients.

5.2 Positive measure and full degree

On the half-square $u > v$, set $u = v + s$. The ranges are $0 < s < 1$ and $0 < v < 1 - s$. Symmetry doubles this integral, and then $y = s^2$ gives $2\,\mathrm{d}s = \mathrm{d}y/s$. Thus μ_m is the mth moment of a measure on $[0, 1]$ with density

$$\rho(y) = \frac{1}{96s} \int_0^{1-s} \frac{\log^2(v(v+s))}{\sqrt{v(v+s)}(1 + v(v+s))}\,\mathrm{d}v, \qquad 0 < y < 1. \tag{5.5}$$

This density is positive throughout $(0, 1)$ and has total mass $\mu_0 = \theta$. Consequently

$$D_n := Q_n^{\Delta}(\theta) = \det[\mu_{i+j}]_{0 \le i,j < n} > 0 \tag{5.6}$$

for every n: a nonzero polynomial has positive squared norm.

Lemma 5.2. *For $n \ge 1$,*

$$\det[C_{i+j}]_{0 \le i,j < n} = 2^{n-1}. \tag{5.7}$$

Thus Q_n^{Δ} has exact degree n and leading coefficient 2^{n-1}.

Proof. The C_m are the moments of the probability arcsine measure $\mathrm{d}y/(\pi\sqrt{y(4-y)})$ on $[0, 4]$. This follows either by the beta integral or by the substitution $y = 2 + 2\cos t$, $0 < t < \pi$, and the binomial expansion of $(2 + 2\cos t)^m$. Let T_j denote the Chebyshev polynomial satisfying $\mathsf{T}_j(\cos t) = \cos(jt)$. The monic integer polynomials

$$c_0(y) = 1, \qquad c_j(y) = 2\mathsf{T}_j((y-2)/2) \quad (j \ge 1)$$

are orthogonal for this measure, with squared norms $1, 2, 2, \ldots$. Monicity and integrality follow from $c_1 = y - 2$, $c_2 = (y-2)^2 - 2$ and the Chebyshev recurrence for $j \ge 2$. The coefficient change from monomials has determinant one. The moment determinant is therefore the product of these norms, giving (5.7). □

5.3 Matching upper and lower real exponents

The upper estimate needs monic polynomials uniformly small on $[0, 1]$; Chebyshev polynomials supply them. For the lower estimate we compare the measure with $(1-y)\,\mathrm{d}y$, whose monic orthogonal norms can be computed exactly. Changing from monomials to any monic polynomial basis uses a triangular matrix with determinant one, so it preserves the Gram determinant in both calculations.

Proposition 5.3. *For all $n \ge 1$,*

$$\frac{1}{6144^n n!\, 2^{2n^2+n}} \le D_n \le \theta^n 4^{-(n-1)^2}. \tag{5.8}$$

In particular,

$$\log D_n = -(\log 4)n^2 + O(n \log(n+1)). \tag{5.9}$$

Proof. For $j \ge 1$ the monic shifted Chebyshev polynomial on $[0, 1]$ has supremum norm $2/4^j$. Change the monomial basis of the Gram matrix to these monic polynomials and apply Hadamard's inequality. Since the measure has total mass θ,

$$D_n \le \theta^n \prod_{j=1}^{n-1} (2/4^j)^2 = \theta^n 4^{-(n-1)^2}.$$

For the lower bound restrict the integral in (5.5) to $v \in [(1-s)/4, (1-s)/2]$. There $v(v+s) \le 1/2$, so $-\log(v(v+s)) \ge \log 2 \ge 1/2$, and the denominator of the integrand is at most 2. The integrand is at least $1/8$, giving

$$\rho(y) \ge \frac{1-s}{3072s} \ge \frac{1-y}{6144} \quad (0 < s < 1). \tag{5.10}$$

The second inequality is $2 \ge s + s^2$. Ordering the positive definite Gram matrices therefore gives

$$D_n \ge 6144^{-n} \det\left[\frac{1}{(i+j+1)(i+j+2)}\right]_{0\le i,j<n} = 6144^{-n} \prod_{j=0}^{n-1} \frac{1}{(2j+2)\binom{2j+1}{j}^2}. \tag{5.11}$$

Here determinant monotonicity follows by conjugating the matrix inequality by the inverse square root of the smaller positive matrix.

We include the norm calculation in the last equality. For weight $1-y$ on $[0,1]$, the monic orthogonal polynomial of degree j is

$$p_j(y) = \frac{(-1)^j (j+1)!}{(2j+1)!(1-y)} \frac{\mathrm{d}^j}{\mathrm{d}y^j}(y^j(1-y)^{j+1}).$$

The numerator is divisible by $1-y$. Integration by parts j times shows orthogonality to degrees below j; every boundary term vanishes. The same calculation with p_j gives its squared norm

$$\frac{j!(j+1)!}{(2j+1)!} \int_0^1 y^j (1-y)^{j+1} \,\mathrm{d}y = \frac{j!^2 (j+1)!^2}{(2j+1)!(2j+2)!} = \frac{1}{(2j+2)\binom{2j+1}{j}^2}.$$

The Gram determinant is the product of these monic norms. Finally $\binom{2j+1}{j} \le 2^{2j+1}$ in (5.11) gives the lower bound in (5.8). Taking logarithms, using $0 < \theta < 1$ and Stirling's formula, proves (5.9). □

6 Exact primitive arithmetic of the difference Gram

Here the prime-denominator contributions first appear at a specific moment index. Because a moment matrix has entry index $i+j$, their residues occupy a corner block with a nonzero anti-diagonal. Its determinant supplies the noncancelling coefficient, just as the single surviving minor did in Remark 4.2. In this family we also bound every coefficient's denominator at the same prime, which turns a lower bound into an exact formula.

Fix $n \ge 1$ and a prime $p > 4n$. All moment indices used in Q_n^{Δ} satisfy $0 \le m \le 2n-2$. In particular $2m < p$, $|j-m| < p$ for the indices in (1.8), and $T_4(m)$ is p-integral. In $T_3(j)$ the only possible odd denominator divisible by p is p itself: the largest odd denominator is at most $8n-9 < 2p$, whereas the next odd multiple is $3p$. It appears when $j \ge (p+1)/2$.

6.1 The first residue and its block

It follows that $p^3 B_m$ is p-integral and

$$p^3 B_m \equiv \frac{\chi_4(p)}{6} \sum_{j=h}^{2m} \binom{2m}{j} \frac{1}{j-m} \pmod{p}, \qquad h = \frac{p+1}{2}. \tag{6.1}$$

An empty sum is zero; in the range of this sum $j > m$, so no zero denominator occurs. Put

$$m_0 = \left\lfloor \frac{p}{4} \right\rfloor + 1, \qquad r = \max\{0, 2n-1-m_0\}. \tag{6.2}$$

Here m_0 is the first moment index for which the sum in (6.1) can be nonempty: $2m \geq (p+1)/2$ is equivalent to $m \geq \lfloor p/4 \rfloor + 1$. For $m < m_0$ no cubic tail contains p, so B_m is actually p-integral. If $m_0 \leq 2n-2$, the first nonzero residue in (6.1), apart from its factor $\chi_4(p)/6$, is

$$\begin{cases} 1/m_0, & p = 4m_0 - 1, \\ (2m_0-1)(m_0+1)/(m_0(m_0-1)), & p = 4m_0 - 3. \end{cases} \tag{6.3}$$

In the first case the sum has only $j = 2m_0$; in the second it has $j = 2m_0 - 1, 2m_0$, giving $2m_0/(m_0-1) + 1/m_0$. These residues are nonzero modulo p. Indeed, in the nonempty case $m_0 \geq 3$, and all the displayed positive numerator and denominator factors are less than p.

If $r = 0$, every residue vanishes. Otherwise, row i can contain a nonzero residue only if $i+j \geq m_0$ for some $j \leq n-1$, hence $i \geq m_0 - (n-1) = n - r$; the same applies to columns. The residue matrix $[p^3 B_{i+j} \bmod p]_{i,j<n}$ is therefore supported in the bottom-right $r \times r$ block. That block is anti-triangular with the nonzero first residue on its anti-diagonal, so it has rank r. More strongly, every entry outside it is p-integral before multiplication by p^3: its moment index is below m_0.

6.2 The exact primitive exponent

Proof of (1.10). At most r entries of a permutation term can have negative valuation, because only the bottom r rows have such entries. Each entry has coefficient valuations at least -3, so every coefficient of Q_n^{Δ} has valuation at least $-3r$.

To see that the bound is attained, suppose $r > 0$, let L be the $r \times r$ residue block just described, and put $s = n - r$. Multiply the bottom r rows of the pencil by p^3. The new matrix has p-integral polynomial entries, and its reduction has the form

$$\begin{pmatrix} [C_{i+j}X + B_{i+j}]_{0 \leq i,j < s} & * \\ 0 & L \end{pmatrix}.$$

The lower-left block vanishes because those entries were already p-integral. In the lower-right block the integral X-coefficients also vanish after multiplication by p^3. Taking this block-triangular determinant proves the polynomial congruence

$$p^{3r} Q_n^{\Delta}(X) \equiv \det(L) \det[C_{i+j}X + B_{i+j}]_{0 \leq i,j < s} \pmod{p}. \tag{6.4}$$

The right-hand coefficient of X^s is $\det(L) 2^{s-1}$ by Lemma 5.2, and is a p-unit. Here $s \geq 1$, since $p > 4n$. Thus the minimum coefficient valuation is exactly $-3r$. If $r = 0$, all entries are p-integral and the leading coefficient 2^{n-1} is already a p-unit. Lemma 2.2 now proves (1.10) in all cases. □

This argument controls the actual primitive multiplier. Any coefficient gcd is already included in the minimum-valuation calculation; it cannot remove the forced exponent.

6.3 Summing the prime cost

By Lemma 2.2 and the leading coefficient 2^{n-1}, $v_p(\kappa_n) \geq 0$ at every odd prime, while $v_2(\kappa_n) \geq -(n-1)$. Hence

$$\log \kappa_n \geq 3 \sum_{4n < p < 8n-8} \left(2n - 2 - \left\lfloor \frac{p}{4} \right\rfloor\right) \log p - (n-1) \log 2. \tag{6.5}$$

Changing the endpoints by a bounded amount and removing floors costs $O(n\log(n+1))$, which is negligible on the quadratic scale. Partial summation and (1.12) give

$$3\sum_{4n<p<8n-8}\left(2n-2-\left\lfloor\frac{p}{4}\right\rfloor\right)\log p = 3\int_{4n}^{8n}(2n-t/4)\,\mathrm{d}\vartheta(t)+O(n\log(n+1))$$
$$= 3\int_{4n}^{8n}(2n-t/4)\,\mathrm{d}t+o(n^2) = 6n^2+o(n^2). \tag{6.6}$$

For the replacement of $\mathrm{d}\vartheta$ by $\mathrm{d}t$, integration by parts bounds the error using $\sup_{4n\le t\le 8n}|\vartheta(t)-t| = o(n)$, the $O(n)$ endpoint weight and its bounded derivative. Thus $\log\kappa_n \ge 6n^2+o(n^2)$. Combining with Proposition 5.3 proves (1.11) and completes the proof of Theorem 1.2.

7 Scope, evidence and the wider investigation

The two proofs establish actual primitive-value obstructions. The first uses a monic pencil and a unique surviving principal minor; the second uses a power-of-two leading coefficient and a full anti-triangular residue block. In both cases the leading coefficient controls possible negative valuations of the primitive multiplier at the primes not used in the lower-bound sum. Without that control, a large denominator cost alone would not prove the same conclusion.

These results concern the full pencils defined in (1.5) and (1.9). Invertible rational basis changes preserve their primitive polynomials, but different subspaces, sparse designs, nonprincipal minors, polynomial weights, factors and combinations of determinants are outside the two obstruction theorems. The uniform real comparison in Theorem 3.3 does apply to nonprincipal designs; an arithmetic obstruction for such a design would require additional information.

The accompanying research catalog separates proved limitations of defined families from failed estimates, exact reductions to existing constructions, finite experiments and incomplete constructions. In particular, some nonprincipal designs have unfavorable proved denominator bounds without an all-index bound on numerator content. An ordinary-moment first-variation construction has a positive error and a proved real exponent, but no all-index primitive arithmetic conclusion. Neither status is upgraded to an impossibility statement by the present theorems. The accompanying technical supplement develops selected further proofs and guides the reader through the remaining constructions, with links to the catalog and source snapshots.

Computational checks and authorship

This note was prepared with Codex. The archived record includes model mathematical cross-checks, exact rational calculations and Arb interval calculations. A reproducibility record accompanies this draft and states which checks were rerun and how the archived outputs were preserved. Such checks are finite evidence for identities and arithmetic formulas; the asymptotic results are proved in the text. No proof-assistant verification or human mathematical audit is claimed. The comparison with prior work is recorded by source and version; no exhaustive priority claim is made.

A Including a common zero node

The unregularized Cauchy matrix is unsuitable when both node lists contain zero, but the regularized matrix in (3.21) remains finite. For $a_0 = b_0 = 0$ and increasing positive remaining nodes, write

a_+, b_+ for the lists with zero removed. Continuity gives $R_\tau(0,0) = 1/\tau$ and

$$\det[R_\tau(a_i, b_j)]_{0\le i,j<n} = \frac{1}{\tau}\prod_{i>0}\frac{a_i}{a_i+\tau}\prod_{j>0}\frac{b_j}{b_j+\tau}\det C(a_+, b_+). \tag{A.1}$$

To prove it, multiply row i by $a_i + \tau$ and column j by $b_j + \tau$. The resulting entry is $\tau + ab/(a+b)$, with continuous value τ in the zero row and column. Subtract the zero row from every other row and expand at the zero column. Undoing the scalings gives (A.1); an empty determinant is one.

The lower density bound (3.18) still applies. For an upper bound finite at zero, note that $1 - \operatorname{sech} v \le v^2/2$ for $v \ge 0$: integrate $\operatorname{sech} v \tanh v \le v$. Therefore

$$q(s) \le 5(1 - \operatorname{sech}(2\pi s)) \le 10\pi^2 s^2, \qquad q(s) \le M.$$

Combining the bounds gives $q(s) \le (M + 10\pi^2\tau^2)s^2/(s^2+\tau^2)$ for every $\tau > 0$. Choose $\tau = \tau_0$; then

$$c_0^n \det R_{\tau_0} \le \det[K(a_i^2, b_j^2)] \le [c_0(M+5)]^n \det R_{\tau_0}. \tag{A.2}$$

For positive integer remaining nodes, (A.1) gives

$$\log\det[K(a_i^2, b_j^2)] = \log\det C(a_+, b_+) + O(n), \tag{A.3}$$

uniformly in the remaining node lists.

Corollary A.1. *For the principal nodes $0, 1, \dots, n-1$, replace θ by X in (3.8)–(3.10), take the determinant, and let Z_n^0 be its primitive representative. Then*

$$\liminf_{n\to\infty}\frac{\log Z_n^0(\theta)}{n^2} \ge 2 - 2\log 2.$$

Proof. The coefficient matrix of X is $\operatorname{diag}(2, 1, \dots, 1)$, so the leading coefficient of the determinant is 2. For primes $2(n-1) < p < 4(n-1)$, the unique high-node minor from Lemma 4.1 is unchanged; its complementary diagonal coefficient contributes a factor 2, a p-unit. Thus the same coefficient valuation $-4r_p$ is attained. All odd prime valuations of the primitive multiplier are nonnegative, and the only possible negative contribution is at least $-\log 2$. The prime sum is $2(n-1)^2 + o(n^2)$. By (A.3), its positive real determinant has logarithm $-2\log 2\,(n-1)^2 + o(n^2)$. Adding proves the assertion. □

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